

Lecture 11

Midterm: - Graded by th.

- Midterm hard $\sim$ normal, median $\geq$ B
- One on one meeting w/GSI $\sim$ email soon.

Feedback Form:

- shorter HW.
- Schedule on bcourses up to MT2.

Plan: ① Induction. ② Graph Theory (4) ③ Counting.

I Induction

So far: Direct Proof, Contrapositive, Contradiction, Cases,
Counterexample $\leadsto \forall x P(x), \exists x P(x)$

New Inference Rule: for proving " $\forall n \in \mathbb{Z}_+ P(n)$ ".

ex 1: $\forall n \in \mathbb{Z}_+ (n! > 2^n)$.

ex 2: $\forall n \in \mathbb{Z}_+ (1+2+\dots+n = \frac{n(n+1)}{2}) \leftarrow$

ex 3: $\forall n \in \mathbb{Z}_+ (\underbrace{3 \mid n^3 - n}_{P(n)})$

Principle of Mathematical Induction: $P(n)$ predicate.

$P(1)$	$\leftarrow$ Base case
$\forall k \in \mathbb{Z}_+ (P(k) \rightarrow P(k+1))$	$\leftarrow$ Inductive Step.
$\therefore \forall n \in \mathbb{Z}_+ P(n)$	

Recall: Summation notation.

final value $\rightarrow$ $\sum_{i=1}^n i = 1 + 2 + 3 + 4 + \dots + n.$

initial value $\rightarrow$

$$\sum_{i=0}^n 2^i = 2^0 + 2^1 + 2^2 + \dots + 2^n.$$

In general: $\sum_{i=a}^b f(i) = f(a) + f(a+1) + \dots + f(b).$

Proposition 1: $\forall n \in \mathbb{Z}_+ \left(\underbrace{\sum_{i=1}^n i = \frac{n(n+1)}{2}}_{P(n)} \right)$.

(ex: $P(5)$: $1+2+3+4+5 = \frac{5 \cdot 6}{2}$.)

Proof: We proceed by induction.

Base Case: Observe that $\sum_{i=1}^1 i = 1 = \frac{1(1+1)}{2}$, so $P(1)$ holds.

Inductive Step: Assume $k \in \mathbb{Z}_+$ is arbitrary.

Assume $P(k)$.

We will show $P(k+1)$.

$\hookrightarrow$
 $P(k)$

Observe that

$$\begin{aligned} \sum_{i=1}^{k+1} i &= \sum_{i=1}^k i + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \end{aligned}$$

Want to show $P(k+1)$ $\leftarrow$ $\xrightarrow{\text{described by } P(k)}$ $\xrightarrow{\text{by } P(k) \text{ inductive}}$

Hypothesis"

$$= (k+1) \left(\frac{k}{2} + 1 \right)$$

$$= \frac{(k+1)(k+2)}{2}$$

$$= \frac{(k+1)((k+1)+1)}{2},$$

establishing $P(k+1)$.

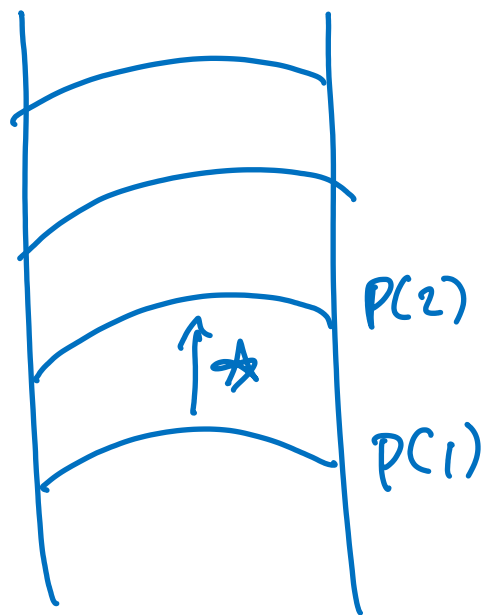
By induction, we conclude that $\forall n \in \mathbb{Z}_+ \left(\sum_{i=1}^n i = \frac{n(n+1)}{2} \right)$. $\square$

Remark: If the base case is b , induction shows $\forall n \geq b (P(n))$.

Logical Structure: $P(1), \underbrace{\forall k \in \mathbb{Z}_+ (P(k) \rightarrow P(k+1))}_{\star}$.

$$P(1) \xrightarrow[\star]{k=1} P(2) \xrightarrow[\star]{k=2} P(3) \rightarrow \dots \rightarrow \dots$$

Metaphor:



Base case: Can step on first rung.
Inductive Step:

Can climb from rung k
to rung $k+1$.

Conclusion: $\forall n \in \mathbb{Z}_+$, you
will eventually reach the n th
rung.

(Q) What about $\forall k \in \mathbb{Z} (P(k+1) \leftrightarrow P(k))$? Does
that imply $\forall n \in \mathbb{Z} (P(n))$?

Remark: Induction cannot prove $\forall n \in \mathbb{R}, (P(n))$.
(no "next" real number)

Validity of induction follows from the well-ordering principle
applied to $S = \{n \in \mathbb{Z}_+ : P(n) \text{ is false}\} \subseteq \mathbb{Z}_+$.

ex 4: For $n \in \mathbb{Z}_+$, define

$$S_n = 1 + 3 + 5 + \dots + (2n-1) = \sum_{i=1}^n (2i-1).$$

$$S_1 = 1, S_2 = 1+3=4, S_3 = 1+3+5=9, S_4 = 1+3+5+7=16$$

these are perfect squares.

Proposition 2: $\forall n \in \mathbb{Z}_+ \quad (\overbrace{\exists m \in \mathbb{Z}_+}^{P(n)} S_n = m^2)$. etronger $S_n = n^2$

Proof: We proceed by induction.

✓ Base Case: $S_1 = 1 = 1^2$ so $P(1)$ holds.

✓ Inductive Step: Assume $k \in \mathbb{Z}_+$ arbitrary. inductive hypothesis Assume $P(k)$.
We will show $P(k+1)$.

$\overset{k}{P(k)}, m$
 $S_k = k^2$

Observe that

$$S_{k+1} = \sum_{i=1}^{k+1} (2i-1) = \underbrace{\sum_{i=1}^k (2i-1)}_{S_k} + (2k+2-1)$$

$$= k^2 + 2k + 1$$

by the inductive
hyp

$$= (k+1)^2,$$

establishing $P(k+1)$.



"inductive loading"

Remark: Sometimes it is easier to prove a
logically stronger statement by induction.

$P(n)$ too weak $\longrightarrow$ inductive step fails.

$P(n)$ too strong $\longrightarrow \forall n$ $P(n)$ is false.

$\therefore$ it is very important to identify the correct $P(n)$.

($A \rightarrow B$ means A is "stronger than" B).

Pol1: $P(1), \underbrace{\forall k \in \mathbb{Z}_+ (P(k) \rightarrow P(k+2))}_{\star}$.

$$P(1) \xrightarrow{k=1} P(3) \xrightarrow{k=3} P(5) \rightarrow P(7) \rightarrow \dots$$

to see this is all, consider $P(n) = "n \text{ is odd}."$

(Q) $P(1), P(2), \star \longrightarrow \forall n \in \mathbb{Z}_+ P(n)$

(Q) Could have P on $\mathbb{Z}_{\geq 0}$, $P(0)$, inductive hyp.
However, wouldn't work for $\forall n \in \mathbb{Z} (P(n))$.
(no base case).

So far: Proofs are about numbers, functions, etc. Now: more general objects.

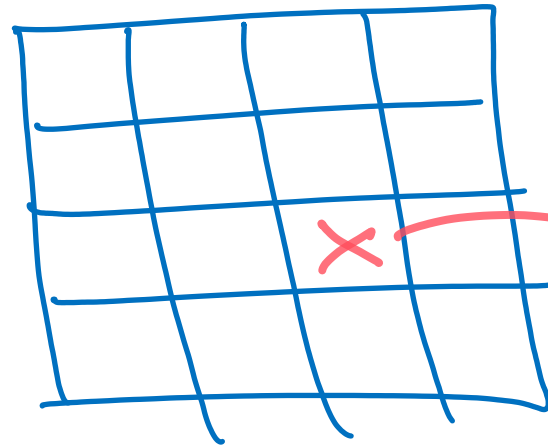
Puzzle: Define a triomino as an element of
 $\left\{ \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \\ \hline \end{array}, \begin{array}{|c|c|} \hline \square & \\ \hline \end{array}, \begin{array}{|c|c|} \hline & \square \\ \hline \end{array} \right\}$

(Q) Can you perfectly tile

every square
covered exactly
once.

a 4×4 chessboard with
triominoes?

with an arbitrary square removed



s .
name of
removed
square.

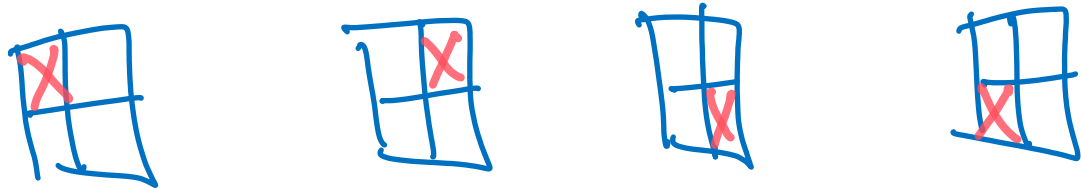
(A) No. 16 squares. $3 \nmid 16$.

Proposition 3: For every $n \in \mathbb{Z}^+$, $\exists$ for every $2^n \times 2^n$ board
with an arbitrary square s removed,
the board can be $\exists$ perfectly tiled with triominoes. $\} P(n)$.

Proof:

By induction.

Base Case: For $n=1$, the board must be one of



each of which can be tiled by a single triomino, so $P(1)$ holds.

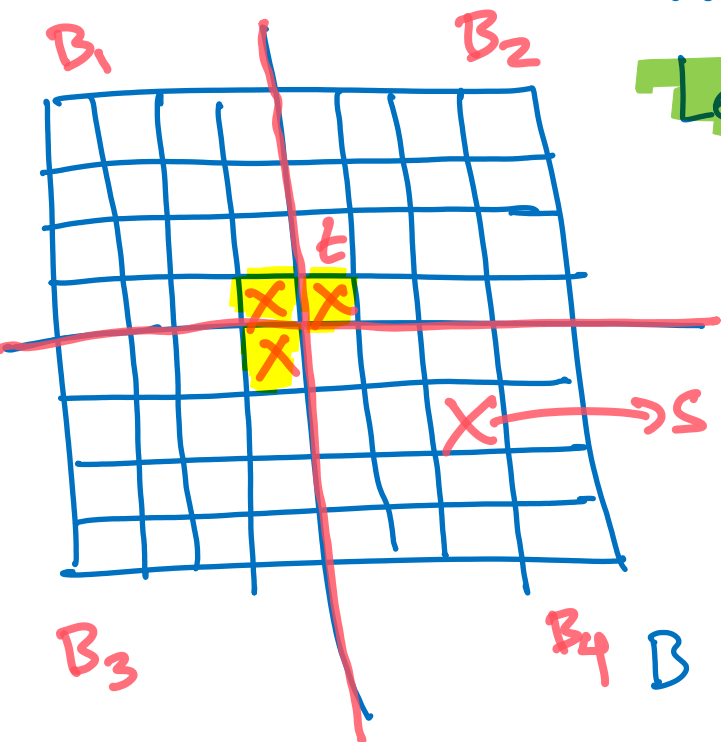
Inductive Step: Assume $k \in \mathbb{Z}_+$ and $P(k)$ is true.

$k=2$

We will show $P(k+1)$.

Let B be a $2^{k+1} \times 2^{k+1}$ board with one square s removed.

Partition B into 4 boards of size $2^k \times 2^k$, call them B_1, B_2, B_3, B_4 as shown in the diagram.



Assume without loss of generality that s lies in B_4 , the bottom right board, by rotating B , if necessary.

Place a triomino t in the center of B with one tile in each of B_1, B_2, B_3 .

Let B'_1, B'_2, B'_3 denote B_1, B_2, B_3 with the squares occupied by t removed.

By PCK, we can choose perfect tilings T_1, T_2, T_3, T_4 of B'_1, B'_2, B'_3, B_4 .

Combining $T_1, \dots, T_4$ with t yields a tiling of B , as desired. $\square$

